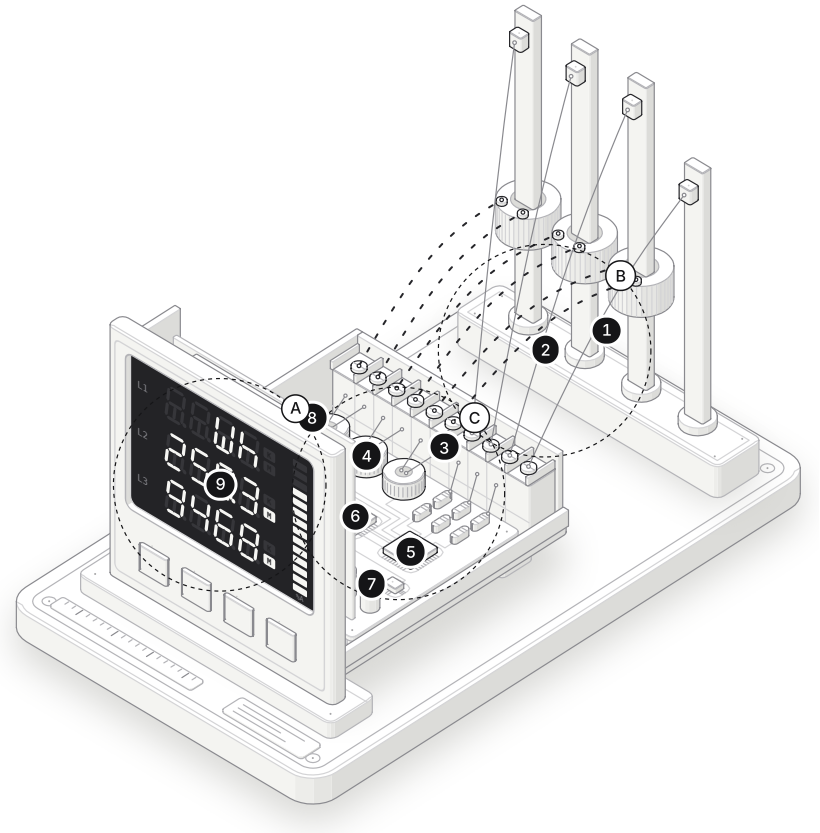
Energy meter



How do you read the units on this meter?

CTs and taps bring each phase's current and voltage in; the metering chip multiplies them into power and the MCU adds it up into Wh. Rows 2 and 3 of the window are one 8-digit number, and the lit K or M lamp gives kWh or MWh.

258.39468 MWh
ON THE WINDOW

2,58,394.68 units
THE SAME IN KWH

400 : 5
CT RATIO

186.3 kW
AT 300 A PER PHASE

PARTS

01 **Bus bars and taps**
L1 L2 L3 carry the load; taps send 230 V.

02 **External CTs 400 : 5**
Step bar current down to 0–5 A.

03 **Terminal block**
I1–I3′ take the CT loops, V1–VN the voltages.

04 **Internal CTs, divider**
Scale current and voltage down for the chip.

05 **Metering IC**
Works out P = V × I × cos φ, three phases.

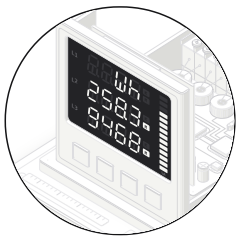
06 **MCU, crystal, EEPROM**
Add power over time into Wh and keep it.

07 **Power supply**
Transformer and capacitors run the meter.

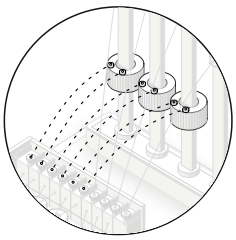
08 **Display board**
Carries the total to the three LED rows.

09 **Window and keys**
Wh row, 8-digit energy, K/M lamps; keys page.

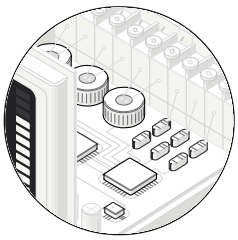
CLOSE-UPS



Ⓐ **The window**
Wh on top; rows 2 and 3 are one number; the lit lamp gives the unit.

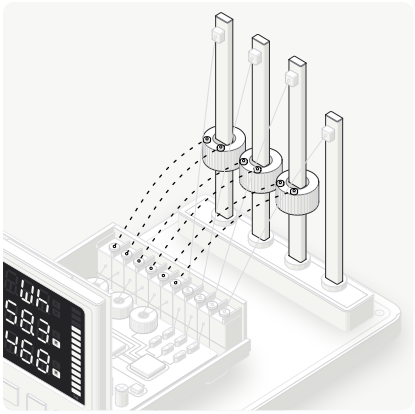


Ⓑ **The CTs**
Each bar passes through a ring CT; S1 and S2 wires run to the meter.



Ⓒ **The board**
Internal CTs and divider feed the metering IC; the MCU counts.

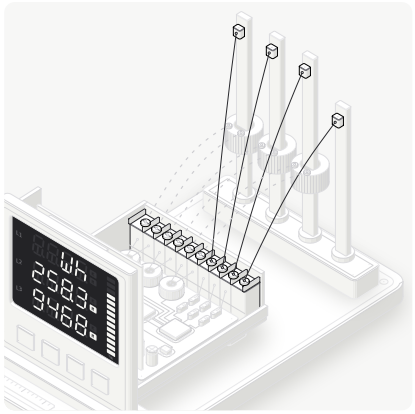
How it works 1 / 2



01

CTs step down

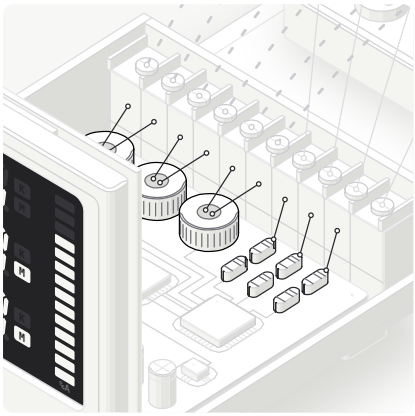
Each bar passes through a 400 : 5 CT, so 300 A in the bar is 3.75 A in its two wires.



02

Voltage tapped

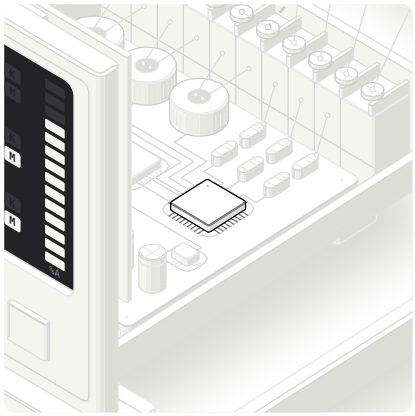
A lug on each bar, and on neutral, sends 230 V to the V terminals.



03

Scaled for the chip

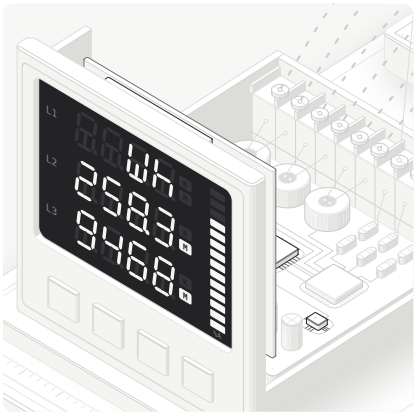
Inside, small CTs take the current down to milliamps and a divider takes 230 V down to a fraction of a volt.



04

Power is worked out

The metering IC multiplies V × I × cos φ for all three phases: 3 × 230 × 300 × 0.9 = 186.3 kW.



05

Energy is counted

The MCU adds power over time into Wh and the EEPROM keeps it through a power cut, so it only goes up.



06

1 · Top row is Wh

Wh is active energy, the billing value. VAh or VARh would not be units.

How it works 2 / 2



07

2 · Read row 2

Exactly as shown, point included: 258.3.



08

3 · Add row 3

Write it straight after: 258.3 and 9468 make 258.39468.



09

4 · K or M lamp

M is lit, so it is 258.39468 MWh. K would mean kWh.



10

5 · Convert to units

MWh × 1000 = 258,394.68 kWh, which is 2,58,394.68 units. Count only bright segments.

WHY TWO ROWS?

Four digits can't hold both a large total and fine steps. Rows 2 and 3 joined make eight digits, so 258.39468 MWh shows the whole total down to 10 Wh, and the lamp says whether it is kWh or MWh.

SPECIFICATION

Supply	3 phase, 4 wire, 230 V L–N	Display	3 rows × 4 digits, K and M lamps
CTs	400 : 5 on the bars	Range	K up to 9999.9999 kWh, then M (figure's rule)
Power factor	0.9 (figure's model)		